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CHINS 100

EQUIPMENT MANUAL

Tortilla Chip Odyssey Oven **CTCO241/217/193/50**

ARTESANO FOODS INC.
SO# 090031.002 IRN# 191117

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NOTE: The following identifiers are used on most of our information sheets as Casa Herrera information file locators:
“CHINS” = C (asa) H (errera) IN (formation) S (heets) (3-digit number)
“MINS” = M (anufacturer’s) IN (formation) S (heets) (4-digit number)

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Mechanical & Electrical Information

Mechanical Drawings & Recommended Spare Parts

- M3774-000102 SHT1 Belt Support & Drives
- M3774-000102 SHT2 Combustion, Infrareads & Pilots
- M3774-000102 SHT3 Combustion Box
- M3774-000102 SHT4 Air Knife Belting & Infeed
- M3774-000102 SHT5 Insulation
- M3774-000102 SHT6 Crumb Trays – Doors & Hood

Electrical Bill of Materials

- 90031.002 Oven Electrical Bill of Materials
- 90031.005 EP-SYST Electrical Bill of Materials
- 90031.006 Cable Tray Electrical Bill of Materials

Electrical Spare Parts

- 90031.002 Oven Electrical Spares
- 90031.005 EP-SYST Electrical Spares

Electrical Drawings

- Refer to 90031.005 EP-SYST-PS-CCFB Elect. Drawings **92056901 - 92056927**

Manufacturer's Information

Sumitomo Gear Motor	MINS 1012
Sumitomo Gear Motor	MINS 1087
Eclipse Blower	MINS 1293
Ashworth Balanced-weave	MINS 0303
AcoTech Burners	MINS 1052
Allen Bradley Panel View	MINS 1080
Allen Bradley, Safety Relay	MINS 1354
Fireye Programming Module.....	MINS 1185
Allen Bradley PowerFlex 525 Series AC Drive	MINS 1331
Allen Bradley Illuminated Push Buttons	MINS 1123
Allen Bradley IEC-Disconnects.....	MINS 1121
ISC Air Conditioner	MINS 1338
Fireye Scanner	MINS 1004
Watlow High Temperature Controller	MINS 1228



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CHINS 101

EQUIPMENT INFORMATION

OVEN, TORTILLA CHIP ODYSSEY

ARTESANO FOODS INC.

S.O. # 090031.002 IRN# 191117

Model No.: **CTCO241/217/193/50**

Serial No.: **OV20031860**

HP/FLA: **3 / 4.8** VOLTS: **460** AMPS/FLA: **25.2** PHASE: **3**

Approx. BTU: **3.9 M**

INSTALLATION DATE: _____

**If you call for service, the above information will
be requested from you!**

MANUFACTURER'S STATEMENT

It is the responsibility of the customer to comply with local and national codes for installation.

Casa Herrera Inc. offers instruction in safe and proper operation of Casa Herrera machinery. Purchasers of our equipment are strongly urged to consider sending their operating and maintenance personnel to our Pomona plant for "hands-on" training and experience.

This product distributed by: **CASA HERRERA, Inc.**
2655 North Pine Street
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MACHINERY RISK ASSESSMENT RECORD

Machinery EHSR's – EU Directive 2006/42/EC
ISO 13849-1 2015

Company: Casa Herrera Inc. **Job #:** 090031_191117 **Machine Type:** Corn Tortilla Oven, Odyssey
Model: CTO241/217/193/50 **Serial #:** OV20031860 **End User:** Artesano Foods **Location:** Austin, TX
Assessor(s): Mike Dale, Ken Shockley & Mark Herrera **Assessment Date:** 11/14/2019 **Rev:** 000
Assessment Stage: 3 (Assessment Stage Key: 1- Design, 2- Partly Assembled, 3- Fully Assembled)

Nature of Hazard:

- 1) Infeed conveyor: Belt Edges. Pinch/Crush Hazard.
- 2) Infeed conveyor: Raising and Lowering into position. Crush/Pinch Hazard.
- 3) Oven: Cooking Chamber. Pinch/Crush Hazards.
- 4) Oven: Cooking Chamber and Surfaces. Burn Hazards.
- 5) Oven: Hood when lowering into position. Crush/Pinch Hazard.
- 6) Oven: Hood Lift Drive, when raising and lowering the hood. Cut/Pinch Hazard.
- 7) Oven: Hood Ends. Crush/Pinch Hazard.
- 8) Discharge conveyor: Belt Edges. Pinch/Crush Hazard.

Safety Assessment Actions and Recommended Safety Procedures:

- 1) The belt edges on the infeed conveyor have a pinch point hazards at the nose bar lock collar location and at the UHMW guide block location. Covers have been added to minimize exposure. Covers require a tool to remove. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 2) The Infeed conveyor is raised and lowered into position manually. It requires 2 operators to raise and lower the discharge conveyor into position because of the position locking pins on either side of the discharge conveyor. There are 3 positions that the discharge conveyor can be locked into. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 3) The oven access doors to the cooking chamber have door latch handles, surfaces may be hot. Doors act as a guard for the hazards in the combustion chamber. A tool is required to open doors. Operators and maintenance personnel must be made aware of the Pinch and Burn Hazards from burners, moving belts, belt supports and tracking rollers when oven is in operation. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 4) There are burn hazards thru out the oven. Refer to note A for safety precautions. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 5) The oven hood can be raised and lowered. When the hood is being lowered into place there is a Pinch/Crush Hazard between the bottom part of the hood and the top part of the oven frame. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 6) The hood lift drive shafts and couplings rotate during operation. There are no guards and although they are out of reach the possibility of injury can occur. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 7) The hood ends can be raised and lowered, creating Crush/Pinch Hazards. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.
- 8) The belt edges on the discharge conveyor have a Pinch/Crush hazards. Covers have been added to minimize exposure. Covers require a tool to remove. Operators and Maintenance Personnel must be made aware of these hazards. Do not bypass or defeat any guarding or safeties.

***A.** Surface temperatures on the Oven may in some areas and will in other areas exceed 110.8 Deg. F (43.8 Deg. C). 110.8 Deg. F (43.8 Deg. C) is the Hot Touch Temperature Limit set for the hand by ISO 13732-1 Part 1 Hot Surfaces. Employer must supply employee proper Personal Protective Equipment based on actual surface temperatures. Reference OSHA 29 CFR 1910.138. Warning labels have been added, where applicable.

HRN: Without guards, covers and safeties.

- 1) $2 \times 2.5 \times 2 \times 1 = 10$, Low to Significant.
- 2) $2 \times 2.5 \times 4 \times 1 = 20$, Significant.
- 3) $5 \times 5 \times 6 \times 1 = 150$, Very High.
- 4) $5 \times 4 \times 4 \times 1 = 80$, High.
- 5) $2 \times 2.5 \times 6 \times 1 = 30$, Significant.

- 6) $1 \times 2.5 \times 0.5 \times 1 = 1.25$, Very Low.
- 7) $2 \times 2.5 \times 4 \times 1 = 20$, Significant.
- 8) $2 \times 2.5 \times 2 \times 1 = 10$, Low to Significant.

HRN: With guards, covers and safeties.

- 1) $2 \times 2.5 \times 0.5 \times 1 = 2.5$, Very Low.
- 2) N/A
- 3) $0.0333 \times 5 \times 2 \times 1 = 0.333$, Negligible.
- 4) $2 \times 4 \times 2 \times 1 = 16$, Significant.
- 5) N/A
- 6) N/A
- 7) N/A
- 8) $2 \times 2.5 \times 0.5 \times 1 = 2.5$, Very Low.

HRN Assessment Key: LO x FE x DPH x NP = HRN

LO (Likelihood of Occurrence, contact with hazard):

0.0333 Almost Impossible - cannot happen

1 Highly Unlikely - though conceivable

1.5 Unlikely - but could occur

2 Possible - but unusual

5 Even chance – could happen

8 Probable - not surprising

10 Likely – only to be expected

15 Certain - no doubt.

FE (Frequency of Exposure, now incorporates Duration):

0.5 Annually

1 Monthly

1.5 Weekly

2.5a Daily – Once to several times per 24 hours, Duration of tasks is Less than or = to 72 minutes total per 24 hours.

2.5b Daily – Once to several times per 12 hours, Duration of tasks is Less than or = to 36 minutes in total.

4 Hourly to Several times per 8 hours, Duration of tasks is less than or = to 24 minutes in total.

5a Daily – Once to several times per 24 hours, Duration of tasks is Greater than 72 minutes total per 24 hours.

5b Daily – Once to several times per 12 hours, Duration of tasks is Greater than 36 minutes in total.

5c Hourly to Several times per 8 hours, Duration of tasks is Greater to 24 minutes in total.

5d Hourly to Daily exposure where the duration of exposure to hazard is $> 1/20^{\text{th}}$ of the operational time.

6 Constantly or Greater than once each 15 minutes.

Note

$1/20^{\text{th}}$ of an 8 hour shift is 24 minutes

$1/20^{\text{th}}$ of a 12 hour shift is 36 minutes

$1/20^{\text{th}}$ of a 24 hour shift is 72 minutes

DPH (Degree of Possible Harm, worst possible case)

0.1 Scratch/Bruise

0.5 Laceration/Mild ill health effect

2 Break of a minor bone, minor illness or minor burn (temporary)

4 Break of major bone, major illness or minor/major burn (temporary)

6 Loss of one limb or eye, hearing, minor/major burn (permanent)

10 Loss of two limbs or eyes, hearing, major burn (permanent)

15 Fatality

NP (Number of Persons at Risk, to the hazard):

1 (1-2 Persons)

2 (3-7 Persons)

4 (8-15 Persons)

8 (16-50 Persons)

12 (16-50 Persons)

HRN (Risk Hazard Rating Number):

0-1 Negligible

1-5 Very Low

5-10 Low

10-50 Significant
50-100 High
100-500 Very High
500-1000 Extreme
1000+ Unacceptable.

Assessment carried out by:

Name: Michael A. Dale

Position: Compliance Tech.

Signature: 

Date: 12/26/2019

Name: Ken Shockley

Position: Project Draftsman

Signature: 

Date: 12/26/19

Name: Mark Herrera

Position: President

Signature: 

Date: 12/30/19



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MANUFACTURER'S STATEMENT

CASA HERRERA gas-fired appliances are designed and built to meet or exceed guidelines set forth by local codes, Factory Mutual (FM), Industrial Risk Insurers (IRI), NFPA 86, or other codes, as specified by the customer.

It is the responsibility of the customer to comply with local and national codes for installation.

CASA HERRERA offers instruction in safe and proper operation of Casa Herrera machinery. Purchasers of our equipment are strongly urged to consider sending their operating and maintenance personnel to our Pomona plant for "hands-on" training. Contact Casa Herrera for technical assistance regarding maintenance and repairs.

NOTICE The purchaser of this unit shall post, in a prominent location, instructions to be followed in the event of odor of natural gas is detected.

**FOR YOUR SAFETY
DO NOT STORE GASOLINE OR OTHER
FLAMMABLE VAPORS OR LIQUIDS IN THE
VICINITY OF THIS OR ANY OTHER APPLIANCE**

WARNING!

Improper installation adjustments alteration service or maintenance can cause property damage, injury or death. Read the installation, operating and maintenance instructions thoroughly before installing or servicing this equipment.

Lighting and shutdown instructions: See "OVEN OPERATION" section, and posted instructions on the oven.

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INFORMATION MANUAL

SAFETY WARNINGS

DO NOT CLEAN, ADJUST OR REPAIR ANY MACHINERY WHILE PLUGGED INTO ELECTRICAL OUTLETS. KEEP HANDS AND ALL FOREIGN OBJECTS CLEAR OF MACHINERY DURING OPERATION.

DANGEROUS HIGH VOLTAGES ARE PRESENT IN ELECTRICAL ENCLOSURES WHEN THE MACHINE IS OPERATING OR WHEN THE ELECTRICAL SAFETY DISCONNECT SWITCH IS ON. DO NOT OPEN ELECTRICAL ENCLOSURES DURING OPERATIONS.

BEFORE STARTING UP, VERIFY THAT ELECTRICAL ENCLOSURES, GUARDS, AND PANELS ARE IN PLACE AND PROPERLY SECURED. IF ANY ELECTRICAL BOX HAS BEEN LEFT OPEN, OR GUARDS OR PANELS ARE MISSING/DAMAGED, DON'T TOUCH ANYTHING! TELL YOUR SUPERVISOR, AND NOTIFY MAINTENANCE SUPERVISION.

CUIDADO

NO LIMPIE, AJUSTE O REPARE CUALQUIER MAQUINARIA QUE ESTE ENCHUFADA. GUARDE NO INTRODUCIR LAS MANOS O CUALQUIER OTRO OBJETO MIENTRAS LA MAQUINARIA ESTA EN ACCION

Safety Guidelines

Always follow your local, state, and federal guidelines.

- ☐ Access to this machine must be restricted to personnel properly trained in all phases of safety, operation, cleaning, and maintenance procedures.
- ☐ Never wear loose-hanging clothing, jewelry, watches, long-sleeved blouses, neckties or other loose items when performing maintenance or operation procedures on this equipment.
- ☐ While in the manufacturing area, all hair must be constrained with netting.
- ☐ **Do Not** - Remove panels while machine is in operation.
- ☐ **Do Not** - Disconnect, remove, jam, or otherwise defeat the operation and purpose of safety devices such as:
 - Limit Switches
 - Door or panel closure switches
 - Safety interlocks
 - Cover-guards
- ☐ **Do Not** - Place or hold anything into areas where parts are moving, not limited to but including, interiors of panel enclosures, chains, sprockets, and belt guards.
- ☐ **Do Not** - Place hands or anything else into the part of the machine where process materials are mixed, fed, worked, or otherwise handled in some manner.
- ☐ **Do Not** - Probe with **ANYTHING** into areas where parts are moving (not limited to, but including interiors of panel enclosures, behind chain/sprocket or belt guards).

CAUTION KEEP FLOORS CLEAN: You must be careful while working around food machinery; the area can become wet and slippery due to, water, oil or dough on the floor.

CAUTION ELECTRICAL HAZARD: Dangerous high voltages are present in electrical enclosures, while the electrical safety disconnects switches are “On”. **Do not open electrical enclosures during operation.** Before start up operation, verify that electrical enclosures are properly secured and all other guards and panels are in place. If any electrical box has been left open, do not touch anything! Notify your maintenance supervisor.

EFFECTS OF ELECTRICAL SHOCK

Current is the killing factor in electrical shock. Voltage is important only in that it determines how much current will flow through a given body resistance. The current necessary to operate a 10-watt light bulb is eight to ten times more current than the amount that would kill a lineman, that is, if it actually breaks through skin and body resistance and flows at this amperage. A pressure of 120 volts is enough to cause a current to flow, which is many times greater than that necessary to kill.

Currents of 100 to 200 milliamperes cause a fatal heart condition known, as ventricular fibrillation for which there is no known remedy. The following figures are given for human resistance to electrical current:

TYPE OF RESISTANCE	RESISTANCE VALUES
Dry Skin.....	100,000 to 600,000 Ohms
Wet Skin	1,000 Ohms
Internal Body	
Hand-to-Foot	400 to 600 Ohms
Ear-To-Ear	About 100 Ohms

With 120 volts and a skin resistance plus internal resistance totaling 1200 Ohms, we would have 1/10-ampere electric current that is 100 milliamperes. If skin contact in the circuit is maintained while the current flows through the skin, the skin resistance gradually decreases. The following is a brief summary of the effects of current.

EFFECTS OF 60-HZ CURRENT ON AN AVERAGE HUMAN

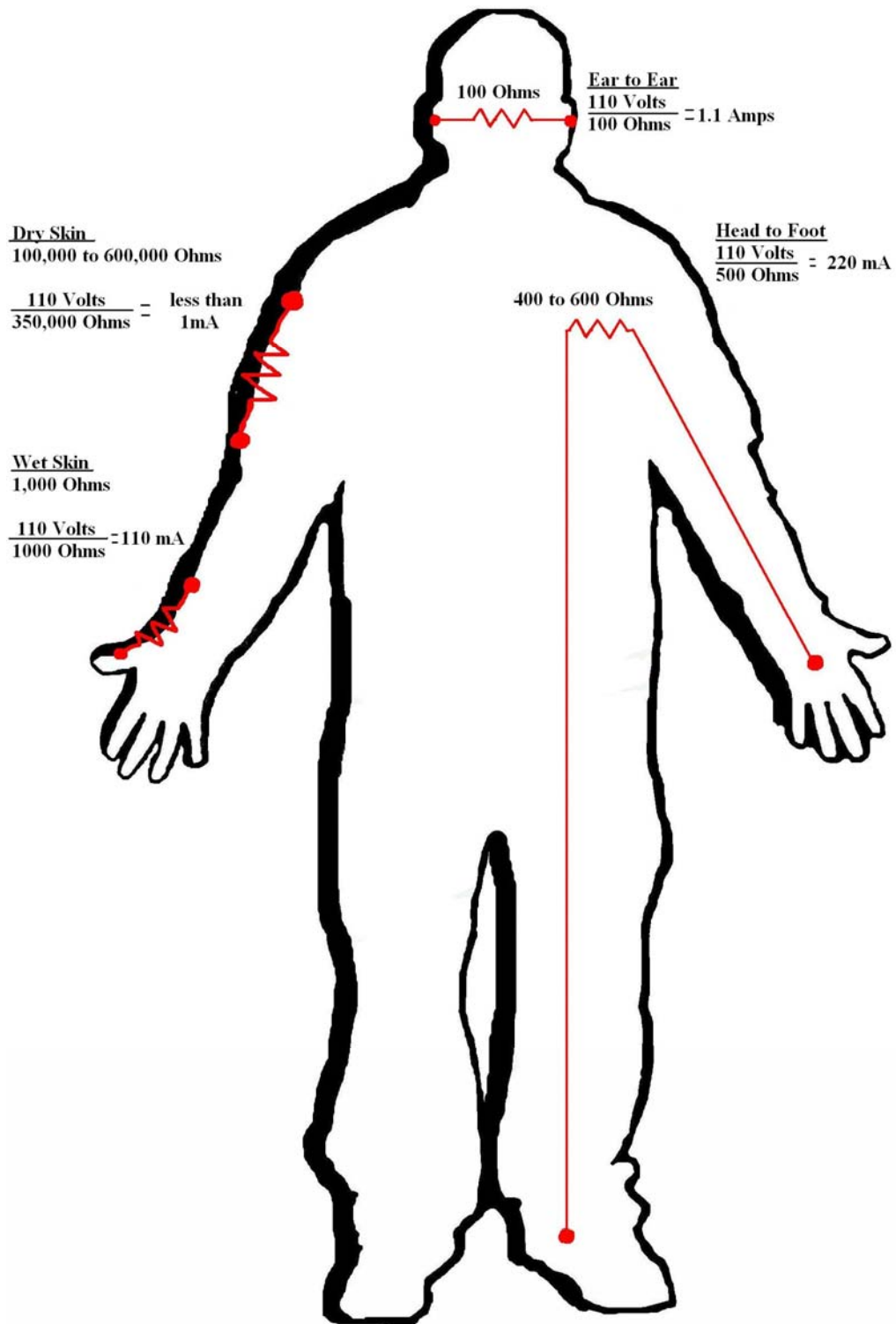
Current Values through Body Trunk		Effect
SAFE CURRENT VALUES	1 Milliampere, or less	CAUSES NO SENSATION - NOT FELT. Is at threshold of perception
	1 to 8 Milliamperes	Sensation of shock. Not painful. Individual can let go at will, as muscular control is not lost. (5mA is accepted as maximum harmless current intensity)
UNSAFE CURRENT VALUES	8 to 15 Milliamperes	Painful shock. Individual can let go at will, as muscular control is not lost
	15 to 20 Milliamperes	Painful shock. Muscular control of adjacent muscles lost. Cannot let go.
	20 to 50 Milliamperers	Painful. Severe muscular contractions. Breathing is difficult
	100 to 200 Milliamperer	VENTRICULAR FIBRILLATION. (A heart condition that results in death - no known remedy)
	200 and over Milliampers	Severe burns. Severe muscular contractions, so severe that chest muscles clamp heart and stop it during duration of shock. (This prevents Ventricular Fibrillation).

*You are your best safety guard

*Keep your resistance high

*Use your safety equipment

*Knowing these hazards is note enough. Take action to be sure you are safe.

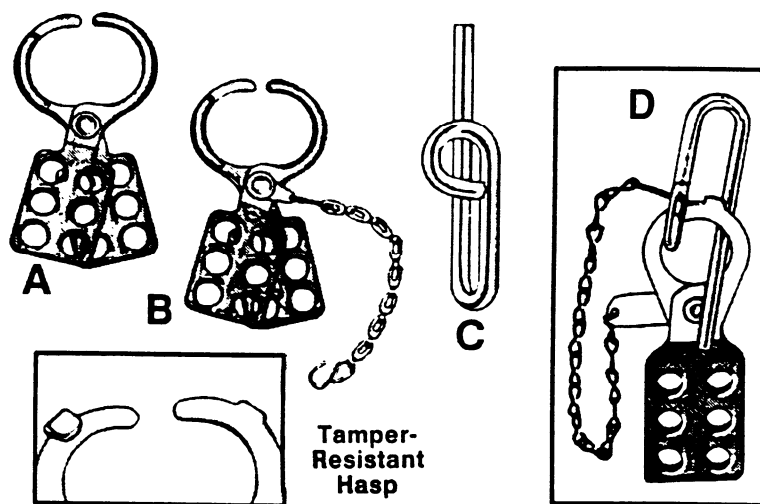


SAFETY LOCKOUT DEVICES

You will see statements in this manual recommending the use of a Safety Lockout Device. These come in different shapes, sizes, and materials, and can usually be purchased locally. The illustration below shows an example, available from McMaster-Carr, PO Box 54960, Los Angeles, CA 90054-0960. It is listed on page 895 of Catalog 97. With a personal padlock and a device like this, an electrical disconnect can be locked in the OFF position to keep a machine from being accidentally turned on.

OSHA regulations now require that lockouts, padlocks, and/or tagouts be used on equipment during repair. Use on electrical switches, circuit breakers, fuse boxes, steam, air and acid valves, controls, levers, and wherever there is a danger to personnel and equipment.

Corrosion-Resistant Lockouts



Red vinyl coating helps prevent corrosion and serves as a warning signal. Each has six holes to accommodate padlocks. Choose from cast aluminum, zinc-plated steel. And stainless steel. Tamper-resistant lockouts have double interlocking tabs on the hasp to help prevent prying hasp open. Chain is for permanent attachment to switches, valves. Fuse boxes. And other control devices. Hasp thickness for all lockouts listed is 1/4" in closed position.